Namal University, Mianwali
Department of Computer Science

CSC-271 – Database Systems
Complex Computing Problem – Milestone 1

Submission Deadline: 12th March, 2026 This milestone is worth **<3%>** of total marks

Instructions:
You have to work in the group; however, marks will be assigned individually based on understanding.

Use of AI tools is allowed for understanding, as long as it is properly referenced. Provide a list of AI used, as appendix to your report.

In this milestone, your goal is to develop a database system proposal that clearly defines the problem, scope, and database requirements, ensuring a solid foundation for the proposed system.

Tasks:

Each group must prepare and submit a proposal for their Database Systems semester project. The proposal should provide a comprehensive plan, outlining key aspects such as problem identification, scope, database requirements, and technological choices.

Deliverable:

Report: A concise report that explains the following:

1. **Introduction:** Briefly introduce the system domain and explain its importance and real-world relevance.
2. **Problem Statement:** Clearly define the specific issue or gap that the proposed database system aims to solve.
3. **Objectives:** State the main goals the system intends to achieve in measurable and clear terms.
4. **Scope:** Define the boundaries of the system by specifying included features.
5. **Data Collection:** Describe the sources and methods used to gather the required system data.
6. **Technology Stack:** List the DBMS and supporting tools or technologies that will be used to develop the system.
7. **System Functionality:** Outline the key functional requirements the system will provide to users. (in numbered points).
8. **User Classes:** Identify different types of users and their respective roles and access levels in the system.
9. **Task Division**: Specify the roles and responsibilities assigned to each team member.
10. **Conclusion:** Summarize the proposal and highlight the expected impact and benefits of the system.
11. **References:** Cite all sources used in proposal development.

12. **AI Prompts (Appendix 1):** Document any AI tools used, including prompts, purposes, and access dates for transparency.

Submission Guidelines:

- The project proposal must be submitted no later than **12th March, 2026 at 02:00 PM.**
- Access the provided sheet link (Click Here) and enter your project title in the designated column.
- Students are required to submit the report in both hard copy and soft copy. Printed and properly stapled/bound report submitted in office and soft copy to be uploaded on QOBE by group lead.
- Upload the complete proposal report to the GitHub repository. Include the GitHub repository link in the References section of the report.

Evaluation Rubric – Database Systems - CCP

	Excellent 1.0	Good 0.7-0.9	Satisfactory 0.5-0.6	Unsatisfactory 0.2-0.4	Poor 0.0-0.1
Viva	Answered all questions correctly	Answered most questions correctly	Answered some questions correctly	Answered very few questions correctly	Answered no questions correctly

		Weight	Excellent 90% - 100%	Good 70% - 89%	Satisfactory 50% - 69%	Unsatisfactory 20% - 49%	Poor < 20%
Project Proposal (CLO 1)	Problem Understanding	5	The domain is clearly and accurately explained, and the problem is specific, realistic, and strongly justified.	The domain is explained clearly, and the problem is mostly defined, though some aspects may lack depth or specificity.	The domain is identified, but the explanation is basic and the problem lacks strong justification or clarity.	The domain explanation is vague, and the problem statement is unclear or weakly connected.	The domain is not properly identified, and the problem statement is either missing or completely irrelevant.
	Objectives & Scope	5	The proposal includes measurable objectives, and the system scope is clearly defined.	Objectives are clear and relevant, but some may be slightly generic, and the scope is mostly defined.	Objectives are present but lack measurability or clarity, and the scope is defined in broad terms without clear boundaries.	Objectives are vague or poorly structured, and the scope is unclear or unrealistic.	Objectives are missing or unrelated to the proposed system, and the scope is undefined or impractical.
	System Requirements & User Roles	6	Functional requirements are comprehensive, user roles are well-defined with appropriate access levels.	Functional requirements are mostly clear and relevant, with minor gaps; user roles are defined but access control may lack detail.	Some functional requirements are listed but lack clarity; user roles are generic or insufficiently explained.	Functional requirements are very limited or unclear, and user roles are poorly defined or inconsistent.	Functional requirements are missing/incorrect, and user roles are not defined or do not reflect the system's requirements.

	Technical Planning	4	Appropriate tools are selected with clear reasoning, and data sources and collection methods are clearly identified and explained.	A suitable DBMS and tools are selected, but justification or explanation of data sources lacks detail.	Technologies are mentioned but without clear reasoning, and data collection methods are vaguely described.	Inappropriate or poorly justified technologies are selected, and data collection planning is unclear.	The selected technologies are inappropriate or not mentioned, and there is no explanation of data sources or system planning.
Professional Conduct & Teamwork (Lab: CLO-3)	Documentation & Professional Presentation	5	The report is professionally structured, properly formatted, and includes accurate academic references and complete AI disclosure.	The report is well organized with minor formatting or referencing issues and mostly complete AI disclosure.	The report has basic structure but lacks professional formatting consistency or complete referencing and AI details.	The report shows poor organization, weak formatting, and incomplete references or AI disclosure.	The report lacks proper structure, contains major formatting issues, and does not include required references or AI disclosure.
	Professional Collaboration & Teamwork	5	Task distribution is clear, contributions are documented, and all members actively participate in the project.	Roles are clearly assigned with minor imbalance, coordination is generally effective, and most members demonstrate active involvement.	Roles are assigned but contribution appears uneven, and coordination among members is limited.	Task distribution is unclear, coordination is weak, and some members show minimal participation or preparedness.	There is no clear evidence of collaboration or task division, and the group demonstrates unprofessional conduct or lack of meaningful contribution.
	GitHub Repository Management	5	Repository is well-structured with meaningful folder organization, clear README file and professional documentation.	Repository is organized with a readable README and reasonable commit activity, with minor structural issues.	Repository exists with basic README and limited organization or commit history.	Repository poorly organized; README incomplete or unclear; minimal commit evidence.	Repository missing, empty, or lacks README and proper structure.

***All marks will be scaled based on Viva/ Understanding**

Note: Appearing for the Viva is mandatory. Failure to appear for the scheduled session will result in a score of 0 for the milestone, regardless of the quality of the submitted report.